

SIMATS ENGINEERING

ASSESSMENT TOOL 2 (APPLICATION BASED PROBLEM SOLVING)

COURSE CODE:CSA09

**COURSE NAME: Programming in
JAVA**

COURSE OUTCOME COVERED

CO5. *Construct database-driven web applications by implementing JDBC connectivity to perform SQL operations such as insert, update, and delete using different statement types. (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 1

Total Marks:50

Annexure A

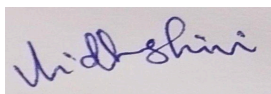
APPLICATION BASED PROBLEM SOLVING

Code of Conduct:

I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

APPLICATION BASED PROBLEM SOLVING

1: Smart Library Book Registration System (Statement + PreparedStatement)

A college library is digitizing its book inventory. The librarian needs a Java application that allows new books to be added while preventing duplicate ISBN numbers from being stored.

Database Table: Books

BookID Title Author ISBN AvailableCopies

Tasks:

1. Establish JDBC connectivity with MySQL.
2. Insert five book records using PreparedStatement.
3. Before insertion, verify whether the ISBN already exists using a Statement.
4. Display an appropriate success or duplicate message.

Problem-solving focus: Choosing between Statement and PreparedStatement for validation and insertion.

2: Hospital Patient Record Updation System (PreparedStatement)

A hospital maintains patient records in a database. Doctors frequently update contact numbers and diagnosis information after each consultation.

Database Table: Patients

PatientID Name Phone Diagnosis LastVisit

Tasks:

1. Connect the Java application to the database using JDBC.
2. Accept a Patient ID from the user.
3. Update the phone number and diagnosis using PreparedStatement.
4. Display the updated patient record after modification.

Problem-solving focus: Secure parameterized update operations.

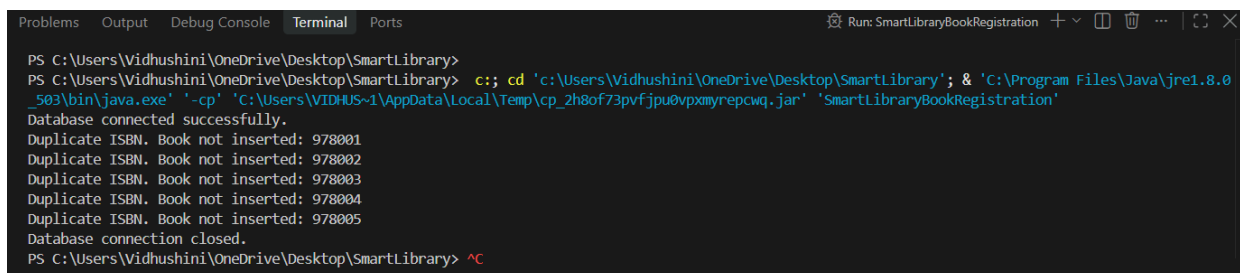
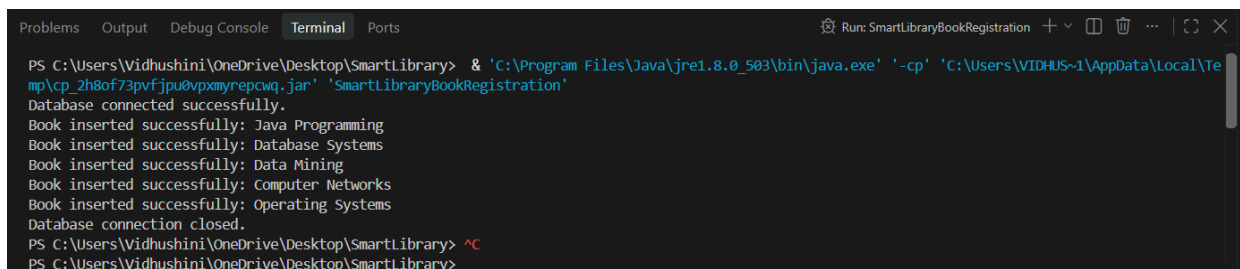
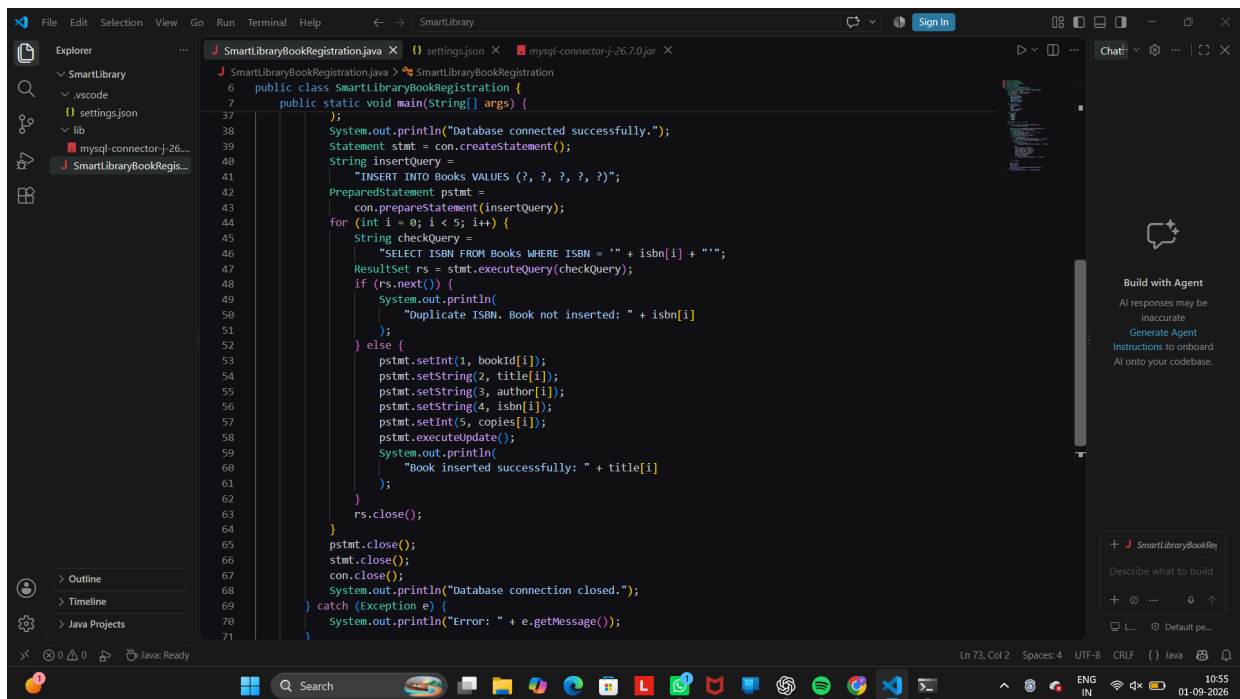
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Analysis & Solution Design	20%	Accurately analyzes the application scenario and designs a complete JDBC-based solution with appropriate SQL operations.	Correctly analyzes the scenario with minor gaps in the solution design.	Partially understands the scenario and proposes an incomplete solution.	Unable to identify the problem requirements or design a suitable solution.
JDBC Connectivity & Statement Usage	25%	Correctly establishes database connectivity and effectively uses the appropriate JDBC statement type (Statement,	Connection and statement implementation are mostly correct with minor errors.	JDBC connectivity is partially implemented; statement usage is inconsistent.	Fails to establish database connectivity or use JDBC statements correctly.

		PreparedStatement, or CallableStatement).			
SQL Operations & Data Integrity	20%	Implements INSERT, UPDATE, and DELETE operations accurately while maintaining data integrity and proper parameter handling.	SQL operations are correct with minor logical or syntax issues.	Performs basic SQL operations with noticeable errors in data handling.	SQL operations are incorrect or fail to modify the database as required.
Program Logic & Exception Handling	20%	Demonstrates strong logical reasoning, validates inputs, and handles exceptions effectively using appropriate techniques.	Good program logic with basic input validation and exception handling.	Limited logical flow with minimal validation or error handling.	Poor program logic with no meaningful exception handling.
Code Quality & Output Verificatio n	15%	Produces clean, modular, well-documented code and correctly verifies database results through meaningful output.	Code is organized with minor documentation or output verification gaps.	Code structure is inconsistent; output verification is incomplete.	Code is poorly structured with incorrect or missing output verification.

```

1  import java.sql.Connection;
2  import java.sql.DriverManager;
3  import java.sql.PreparedStatement;
4  import java.sql.ResultSet;
5  import java.sql.Statement;
6  public class SmartLibraryBookRegistration {
7
8      Run Debug
9      public static void main(String[] args) {
10         String url = "jdbc:mysql://localhost:3306/librarydb";
11         String username = "root";
12         String password = "widhushini2007!";
13         int[] bookid = {1, 2, 3, 4, 5};
14         String[] title = {
15             "Java Programming",
16             "Database Systems",
17             "Data Mining",
18             "Computer Networks",
19             "Operating Systems"
20         };
21         String[] author = {
22             "James Gosling",
23             "Korth",
24             "Han",
25             "Andrew Tanenbaum",
26             "Silberschatz"
27         };
28         String[] isbn = {
29             "978001",
30             "978002",
31             "978003",
32             "978004",
33             "978005"
34         };
35         int[] copies = {5, 3, 4, 6, 2};
36         try {
37             Connection con = DriverManager.getConnection(
38                 url, username, password
39             );

```



```
mysql> CREATE DATABASE librarydb;
Query OK, 1 row affected (0.01 sec)

mysql> USE librarydb;
Database changed
mysql> CREATE TABLE Books (
  ->   BookID INT PRIMARY KEY,
  ->   Title VARCHAR(100),
  ->   Author VARCHAR(100),
  ->   ISBN VARCHAR(20),
  ->   AvailableCopies INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> ALTER TABLE Books
  -> ADD CONSTRAINT unique_isbn UNIQUE (ISBN);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC Books;
```

Field	Type	Null	Key	Default	Extra
BookID	int	NO	PRI	NULL	
Title	varchar(100)	YES		NULL	
Author	varchar(100)	YES		NULL	
ISBN	varchar(20)	YES	UNI	NULL	
AvailableCopies	int	YES		NULL	

```
mysql> SELECT * FROM Books;
Empty set (0.00 sec)

mysql> USE librarydb;
Database changed
mysql> SELECT * FROM Books;
```

BookID	Title	Author	ISBN	AvailableCopies
1	Java Programming	James Gosling	978001	5
2	Database Systems	Korth	978002	3
3	Data Mining	Han	978003	4
4	Computer Networks	Andrew Tanenbaum	978004	6
5	Operating Systems	Silberschatz	978005	2

```
mysql>
```